

Patient Details

PATIENT: XXXXXXXXXX

DATE OF BIRTH: XXXXXXXXXX

ORDERING CLINICIAN: Dr Swapna Patil

SAMPLE TYPE: Buccal

ASSAY ORDERED: POD Health PGx

SAMPLE ID: 1325006529

SAMPLE COLLECTION DATE: 05/06/2025

REPORT DATE: 16/08/2025, 05:59 PM

POD Health, Inc.: Empowering Medication Management with Genetic Insights. At POD Health, we are committed to advancing the precision of mental health treatment through cutting-edge genetic insights. Our PGx Reports are designed to provide healthcare professionals with valuable information to enhance medication management for individuals with mental health conditions and related co-morbidities.

Each POD Health PGx Report includes direct contact information for scheduling consultations with our expert psycho-pharmacologists. These complimentary consultations offer personalized support, helping clinicians make informed decisions about prescribing medications. Our goal is to optimize treatment strategies by leveraging genetic data, ensuring that each patient receives the most effective and tailored care possible.

The POD Health PGx Report serves as a critical tool in understanding genetic factors that can influence medication response. By integrating these insights into treatment planning, healthcare professionals can improve outcomes and provide more individualized care for their patients.

Therapeutic Response annotations

- Good response
- Average response
- Toxicity
- Poor Response
- No interaction detected.
- Use as per recommendation

Metabolizer Phenotypes

- ▲ Ultrarapid Metabolizer
- ▲ Rapid Metabolizer
- ▲ Normal Metabolizer
- ▲ Intermediate Metabolizer
- ▲ Poor Metabolizer

Drug Transporter Type

- △ Increased function
- △ Normal Function
- △ Decreased function
- △ No function
- △ Unknown function

FDA & CPIC Guideline annotation

- ★ : FDA
- ★ : CPIC

The pharmacogenomic (PGx) information in this report is based on peer-reviewed research and the patient's unique genetic profile. It helps assess the patient's likely response to treatment and risk of side effects. Functional and phenotypic insights were inferred via specialized computational frameworks, such as PharmCAT and Stargazer.

Electronically Signed By _____

Contact Information _____

Literature Information Reviewed By _____

Phone number _____

Comment _____

Email _____

GENE DRUG INTERACTION

SUMMARY

Antidepressant

Serotonin ○

SSRI

- Citalopram (Celexa®) ●△
- Fluoxetine (Prozac®) ●
- Paroxetine (Paxil®) ●
- Sertraline (Zoloft®) ●▲
- Escitalopram (Lexapro®) ●△
- Fluvoxamine (Luvox®) ○

SNRI

- Venlafaxine (Effexor®) ●
- Desvenlafaxine (Pristiq®) ●
- Duloxetine (Cymbalta®) ●
- Milnacipran (Savella®) ○

NDRI

- Bupropion (Wellbutrin®) ●

NaSSA

- Mirtazapine (Remeron®) ●

SARI

- Nefazodone ●

TCA

- Clomipramine (Anafranil®) ●
- Nortriptyline (Pamelor®) ●△
- Desipramine (Norpramin®) ●
- Amitriptyline (Elavil®) ○
- Trimipramine (Surmontil®) ○
- Doxepin (Sinequan®) ○
- Imipramine (Tofranil®) ○
- Maprotiline (Ludiomil®) ○
- Opipramol ○

Atypical

- Tianeptine ○

Selective Serotonin Reuptake Inhibitors

- Desmethylcitalopram ○

Atypical antipsychotic

- Clozapine (Clozaril®) ●▲
- Olanzapine (Zyprexa®) ●△
- Risperidone (Risperdal®) ●△
- Quetiapine (Seroquel®) ●
- Aripiprazole (Abilify®) ●△
- Iloperidone (Fanapt®) ●
- Amisulpride ○
- Ziprasidone (Geodon®) ○
- Paliperidone (Invega®) ○
- Lurasidone (Latuda®) ○
- Nemonapride ○
- N-desmethyclozapine ○

Non-stimulant ADHD medication

NRI

- Atomoxetine (Strattera®) ●

Typical antipsychotic

- Chlorpromazine (Thorazine®) ●
- Trifluoperazine (Stelazine®) ●
- Perphenazine (Trilafon®) ○
- Flupenthixol ○
- Bromperidol ○
- Levomepromazine ○

Butyrophenone

- Haloperidol (Haldol®) ●

Thioxanthene

- Zuclopenthixol ○

Mood stabilizer

- Lithium (Lithobid®) ●

Opioid analgesic

Depressant

- Morphine (MS Contin®) ●

Opioid antagonist

- Naltrexone (ReVia®) ○

Statin

- Atorvastatin (Lipitor®) ●△
- Simvastatin (Zocor®) ○
- Rosuvastatin (Crestor®) ○
- Pravastatin (Pravachol®) ○
- Lovastatin (Mevacor®) ○

Beta-blocker

Non-selective, Alpha-1 Blocker

- Carvedilol (Coreg®) ○

Calcium channel blocker

- Diltiazem (Cardizem®) ○
- Nicardipine (Cardene®) ○
- Nimodipine (Nimotop®) ○

Antidiabetic

GLP-1 Receptor Agonist

- Dulaglutide (Trulicity®) ○

Anticoagulant

- Warfarin (Coumadin®) ○

Proton pump inhibitor

- Lansoprazole (Prevacid®) ○

Loop diuretic

- Furosemide (Lasix®) ○

Cholesterol absorption inhibitor

- Ezetimibe (Zetia®) ○

Thyroid hormone replacement

- Liothyronine (Cytomel®) ●
- Levothyroxine (Synthroid®) ○

Disclaimer: This report is intended to supplement a comprehensive patient assessment, including but not limited to diagnosis, clinical history, and evaluation of co-morbid conditions. It should not be used as the sole basis for prescribing decisions. For the most current information on gene-drug associations, please refer to resources such as CPIC, FDA Table of Pharmacogenetic Associations. Please note that pharmacokinetics and pharmacodynamics may evolve over time, and this report may not reflect the latest findings. Prescribers should also consider approved drug labeling, clinical practice guidelines, and other relevant factors such as family history when making treatment decisions.

GENE DRUG INTERACTION

SUMMARY

Stimulant

- Amphetamine
- Dextroamphetamine (Dexedrine®) ●
- CNS Stimulant
- Methylphenidate (Ritalin®) ●
- Nicotinic Receptor Agonist
- Nicotine (Nicorette®) ○
- Xanthine Derivative
- Caffeine (NoDoz®) ○

Benzodiazepine

- Depressant
- Lorazepam (Ativan®) ○
- Diazepam (Valium®) ○
- Alprazolam (Xanax®) ○
- Phenazepam ○

Antiepileptic

- Ethosuximide (Zarontin®) ●
- Levetiracetam (Keppra®) ●
- Oxcarbazepine (Trileptal®) ●
- Cannabidiol (Epidiolex®) ●
- Topiramate (Topamax®) ○
- Lacosamide (Vimpat®) ○
- Mood Stabilizer
- Valproic acid (Depakote®) ●▲
- Carbamazepine (Tegretol®) ●△
- Lamotrigine (Lamictal®) ●△

GABA Analogue

- Gabapentin (Neurontin®) ○

Hydantoin

- Phenytoin (Dilantin®) ●

Benzodiazepine

- Clobazam (Onfi®) ●△

Barbiturate Derivative

- Primidone (Mysoline®) ●

Barbiturate, Depressant

- Phenobarbital (Luminal®) ●△

Anticonvulsant

- Perampanel (Fycompa®) ●▲

Antiviral

- NNRTI
- Efavirenz (Sustiva®) ○
- Neuraminidase Inhibitor
- Oseltamivir (Tamiflu®) ○
- Nucleoside Analogue
- Ribavirin (Copegus®) ●

Immunomodulator

- Interferon
- Peginterferon alfa-2b (PegIntron®) ●
- Peginterferon alfa-2a (Pegasys®) ○

Dissociative anesthetic

- Ketamine (Spravato®) ○

Nutritional supplement

- L-methylfolate (Deplin®) ○

SERM

- Raloxifene (Evista®) ○
- Tamoxifen (Nolvadex®) ○

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SSRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Citalopram (Celexa®)	CREB1 (rs7569963) G>A (AA)	This patient's genetic profile suggests a reduced risk-involved response to citalopram; the 16-item Quick Inventory of Depressive Symptomatology–Clinician Rated, which measures depressive symptoms, was used to assess treatment response.		PMID:17548750
Citalopram (Celexa®)	METTL21A (rs7569963) G>A (AA)	This patient's genetic profile suggests a reduced risk-involved response to citalopram; the Quick Inventory of Depressive Symptomatology (QIDS-C), which measures the severity of depressive symptoms, was used to assess treatment response.		PMID:17548750
Citalopram * (Celexa®)	GSK3B (rs334558) A>G (AG)	This patient's genetic profile suggests a reduced response to citalopram; the 21-item Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to evaluate treatment response.		PMID:18195729
Fluoxetine * (Prozac®)	GSK3B (rs334558) A>G (AG)	This patient's genetic profile suggests a reduced response to fluoxetine; the 21-item Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to evaluate treatment response.		PMID:18195729
Paroxetine * (Paxil®)	GDNF (rs2216711) G>A (AA)	This patient's genetic profile suggests a reduced response to paroxetine; the Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to assess treatment response.		PMID:24577123
Citalopram (Celexa®)	SRP19 (rs495794) A>G (GG)	This patient's genetic profile suggests a better response to citalopram; the 17-item Hamilton Depression Rating Scale (HAMD-17), which measures depression severity, was used to evaluate treatment response.		PMID:22795047
Fluoxetine (Prozac®)	SRP19 (rs495794) A>G (GG)	This patient's genetic profile suggests a better response to fluoxetine; the 17-item Hamilton Depression Rating Scale (HAMD-17), which measures depression severity, was used to evaluate treatment response.		PMID:22795047
Paroxetine (Paxil®)	SRP19 (rs495794) A>G (GG)	This patient's genetic profile suggests a better response to paroxetine; the 17-item Hamilton Depression Rating Scale (HAMD-17), which measures depression severity, was used to evaluate treatment response.		PMID:22795047
Sertraline (Zoloft®)	SRP19 (rs495794) A>G (GG)	This patient's genetic profile suggests a better response to sertraline; the 17-item Hamilton Depression Rating Scale (HAMD-17), which measures depression severity, was used to evaluate treatment response.		PMID:22795047

Disclaimer: This report is intended to supplement a comprehensive patient assessment, including but not limited to diagnosis, clinical history, and evaluation of co-morbid conditions. It should not be used as the sole basis for prescribing decisions. For the most current information on gene-drug associations, please refer to resources such as CPIC, FDA Table of Pharmacogenetic Associations. Please note that pharmacokinetics and pharmacodynamics may evolve over time, and this report may not reflect the latest findings. Prescribers should also consider approved drug labeling, clinical practice guidelines, and other relevant factors such as family history when making treatment decisions.

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SSRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Citalopram * (Celexa®)	FKBP5 (rs4713916) A>G (AG)	This patient's genetic profile suggests a better response to citalopram; the Quick Inventory of Depressive Symptomatology (QIDS-C16), which measures depression severity, was used to evaluate treatment response.		PMCID:PMC2587308
Citalopram (Celexa®)	HTR1B (rs6296) C>G (CG)	This patient's genetic profile suggests a reduced risk of agitation when treated with citalopram; the SSRI Side effect Profile Inventory (SEPI), which measures side effects associated with selective serotonin reuptake inhibitors, was used to assess treatment tolerability.		PMID:27324805
Citalopram (Celexa®)	HTR1B (rs6296) C>G (CG)	This patient's genetic profile suggests a reduced risk of agitation when treated with citalopram; the SSRI Side effect Profile Inventory (SEPI), which measures side effects associated with selective serotonin reuptake inhibitors, was used to assess treatment tolerability.		PMID:27324805
Paroxetine (Paxil®)	HTR1B (rs130058) T>A (TA)	This patient's genetic profile suggests an increased risk of experiencing suicidal thoughts during paroxetine treatment; the 10th item on the Montgomery-Åsberg Depression Rating Scale (MADRS), which assesses suicidal ideation, was used to evaluate treatment response.		PMID:21449676
Escitalopram (Lexapro®)	HTR1B (rs11568817) A>C (AC)	This patient's genetic profile suggests an increased risk of adverse cognitive effects when treated with escitalopram; the digit span test, which measures attention and working memory, was used to assess cognitive function.		PMCID:PMC3967864
Citalopram * (Celexa®)	CRHR2 (rs2270007) G>C (GC)	This patient's genetic profile suggests a reduced response to citalopram; the 21-item Hamilton Depression Rating Scale (HDRS), which measures depression severity, was used to evaluate treatment response.		PMID:17467808
Citalopram * (Celexa®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a better response to citalopram; the Quick Inventory of Depressive Symptomatology Self-Report (QIDS-SR), which measures the severity of depressive symptoms, was used to assess treatment response.	 	PMCID:PMC2268970
Escitalopram * (Lexapro®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced response to escitalopram; the measure of response was used to assess treatment efficacy.	 	PMID:36681895

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SSRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Citalopram * (Celexa®)	SERPINE1 (rs2227631) A>G (GG)	This patient's genetic profile suggests a reduced response to citalopram; the 21-item Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to evaluate treatment response.	●	PMID:18794724
Fluoxetine * (Prozac®)	SERPINE1 (rs2227631) A>G (GG)	This patient's genetic profile suggests a reduced response to fluoxetine; the 21-item Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to assess treatment response.	●	PMID:18794724
Paroxetine (Paxil®)	HTR7 (rs7905446) T>G (GG)	This patient's genetic profile suggests a better response to paroxetine; the 21-item Hamilton Depression Rating Scale (HDRS-21), which measures depression severity, was used to assess treatment response.	●	PMID:30874608
Fluoxetine (Prozac®)	HTR7 (rs7905446) T>G (GG)	This patient's genetic profile suggests a better response to fluoxetine; the 21-item Hamilton Depression Rating Scale (HDRS-21), which measures depression severity, was used to evaluate treatment response.	●	PMID:30874608
Citalopram (Celexa®)	CYP2C19 (rs4244285) G>A (GA)	This patient's genetic profile suggests a moderate response to citalopram; the 21-item Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to evaluate treatment response.	●	PMID:33608663
Paroxetine * (Paxil®)	BDNF (rs6265) C>T (CT)	This patient's genetic profile suggests a reduced response to paroxetine; the Hamilton Depression Rating Scale (HAM-D), which measures depression severity, was used to assess treatment response.	●	PMID:24577123
Escitalopram * (Lexapro®)	BDNF (rs962369) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation during treatment with escitalopram; the Hamilton Depression Rating Scale (HDRS-17), Beck Depression Inventory (BDI), and Montgomery-Åsberg Depression Rating Scale (MADRS), which measure depression severity and suicidal thoughts, were used to assess treatment response.	●	PMID:19641488
Fluoxetine (Prozac®)	HTR2A (rs6313) G>A (AA)	This patient's genetic profile suggests a reduced response to fluoxetine; the Clinical Global Impressions-Improvement score (CGI-I), which measures treatment improvement, was used to assess treatment response.	●	PMID:32073889

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SSRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Escitalopram (Lexapro®)	HTR2A (rs6311) C>T (TT)	This patient's genetic profile suggests an increased risk of adverse cognitive effects when treated with escitalopram; the digit span test, which measures attention and working memory, was used to assess cognitive function.		PMCID:PMC3967864
Citalopram (Celexa®)	HTR2A (rs6311) C>T (TT)	This patient's genetic profile suggests a reduced risk of sexual dysfunction associated with citalopram; the Changes in Sexual Functioning Questionnaire (CSFQ), which measures sexual functioning, was used to assess potential side effects.		PMID:16710319
Escitalopram (Lexapro®)	HTR2A (rs6311) C>T (TT)	This patient's genetic profile suggests a reduced risk of sexual dysfunction when treated with escitalopram; the Changes in Sexual Functioning Questionnaire (CSFQ), which measures sexual functioning, was used to assess treatment-related side effects.		PMID:16710319
Fluoxetine (Prozac®)	HTR2A (rs6311) C>T (TT)	This patient's genetic profile suggests a reduced risk of sexual dysfunction when treated with fluoxetine; the Changes in Sexual Functioning Questionnaire (CSFQ), which measures sexual functioning, was used to assess treatment effects.		PMID:16710319
Paroxetine (Paxil®)	HTR2A (rs6311) C>T (TT)	This patient's genetic profile suggests a reduced risk of sexual dysfunction when treated with paroxetine; the Changes in Sexual Functioning Questionnaire (CSFQ), which measures sexual functioning, was used to assess potential side effects.		PMID:16710319
Citalopram (Celexa®)	NEDD4L (rs520210) G>A (AA)	This patient's genetic profile suggests a better response to citalopram; the 16-item Quick Inventory of Depressive Symptomatology (QIDS-SR), which measures depressive symptoms, was used to assess treatment response.		PMID:23809733
Escitalopram (Lexapro®)	IL11 (rs1126757) C>T (TT)	This patient's genetic profile suggests a better response to escitalopram; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to evaluate treatment response.		PMID:23142150
Citalopram (Celexa®)	GRIA3 (rs4825476) G>A (A)	This patient's genetic profile suggests a reduced risk of suicidal ideation when treated with citalopram; allele and genotype frequencies were used to assess the potential for treatment-emergent suicidal ideation.		PMID:17898344
Sertraline  (Zoloft®)	CYP2B6 *1/*6	This patient's genetic profile suggests an increased risk of adverse drug reactions to sertraline; plasma concentrations measured by high-performance liquid chromatography were used to assess treatment response.		PMID:26830411

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SNRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Venlafaxine (<i>Effexor®</i>)	FKBP5 (rs1360780) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation when treated with venlafaxine; the Montgomery–Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676
Venlafaxine (<i>Effexor®</i>)	HTR1B (rs130058) T>A (TA)	This patient's genetic profile suggests a better response to venlafaxine; the 10th item on the Montgomery–Åsberg Depression Rating Scale (MADRS), which measures suicidal ideation, was used to assess treatment response.		PMID:21449676
Desvenlafaxine (<i>Pristiq®</i>)	HTR7 (rs7905446) T>G (GG)	This patient's genetic profile suggests a better response to desvenlafaxine; the 21-item Hamilton Depression Rating Scale (HDRS-21), which measures depression severity, was used to assess treatment response.		PMID:30874608
Duloxetine (<i>Cymbalta®</i>)	HTR7 (rs7905446) T>G (GG)	This patient's genetic profile suggests a better response to duloxetine; the 21-item Hamilton Depression Rating Scale (HDRS-21), which measures depression severity, was used to evaluate treatment response.		PMID:30874608
Venlafaxine (<i>Effexor®</i>)	HTR7 (rs7905446) T>G (GG)	This patient's genetic profile suggests a better response to venlafaxine; the 21-item Hamilton Depression Rating Scale (HDRS-21), which measures depression severity, was used to evaluate treatment response.		PMID:30874608
Duloxetine (<i>Cymbalta®</i>)	NCAM1 (rs2303377) T>C (TC)	This patient's genetic profile suggests a reduced response to duloxetine; the percentage change in Montgomery – Åsberg Depression Rating Scale score, which measures depression severity, was used to evaluate treatment response.		PMID:28696415
Venlafaxine (<i>Effexor®</i>)	HTR2A (rs7997012) A>G (AG)	This patient's genetic profile suggests a better response to venlafaxine; the Hamilton Anxiety Scale (HAM-A), which measures anxiety severity, was used to assess treatment response.		PMID:22006095
Venlafaxine * (<i>Effexor®</i>)	SLC6A2 (rs2242446) C>T (TT)	This patient's genetic profile suggests a reduced response to venlafaxine; the Montgomery–Åsberg Depression Rating Scale (MADRS), which measures the severity of depressive symptoms, was used to assess treatment response.		PMID:28068779
Venlafaxine (<i>Effexor®</i>)	SLC6A4 (rs25531) T>C (TC)	This patient's genetic profile suggests a reduced response to venlafaxine; the Hamilton Anxiety Scale (HAM-A), which measures anxiety severity, was used to assess treatment response.		PMID:22907732

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SNRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Venlafaxine * (Effexor®)	GABRQ (rs3810651) A>T (T)	This patient's genetic profile suggests a reduced response to venlafaxine; the 17-item Hamilton Depression Rating Scale (HDRS-17), which measures depression severity, was used to evaluate treatment response.		PMID:23394390

NDRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Bupropion * (Wellbutrin®)	SLC18A2 (rs363226) G>C (CC)	This patient's genetic profile suggests a reduced response to bupropion; the 17-item Hamilton Depression Rating Scale (HAMD 17), which measures depression severity, was used to evaluate treatment response.		PMID:22947179
Bupropion * (Wellbutrin®)	HTR2A (rs2770296) C>T (TT)	This patient's genetic profile suggests a reduced response to bupropion; the Hamilton Rating Scale for Depression (HAMD 17), which measures depression severity, was used to evaluate treatment response.		PMID:22947179

NaSSA

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Mirtazapine * (Remeron®)	GAL (rs948854) C>T (TT)	This patient's genetic profile suggests a better response to mirtazapine; the Hamilton Depression Rating Scale (HAMD), which measures depression severity, was used to evaluate treatment response.		PMCID:PMC3055457

SARI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Nefazodone	FKBP5 (rs1360780) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation during treatment with nefazodone; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

SARI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Nefazodone	HTR1B (rs130058) T>A (TA)	This patient's genetic profile suggests a better response to nefazodone; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures the severity of depressive symptoms, was used to assess treatment response.		PMID:21449676
Nefazodone	ABCB1 (rs2032582) A>C (CC)	This patient's genetic profile suggests a reduced risk of suicidal ideation when treated with nefazodone; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676

TCA

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Clomipramine (Anafranil®)	FKBP5 (rs1360780) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation when treated with clomipramine; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676
Clomipramine (Anafranil®)	HTR1B (rs130058) T>A (TA)	This patient's genetic profile suggests a better response to clomipramine; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676
Nortriptyline * (Pamelor®)	UST (rs2500535) A>G (GG)	This patient's genetic profile suggests a better response to nortriptyline; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures the severity of depressive symptoms, was used to assess treatment response.		PMID:20360315
Nortriptyline * (Pamelor®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced risk of hypotension when treated with nortriptyline; the Hamilton Depression Rating Scale (HDRS) and Montgomery & Asberg Depression Rating Scale (MADRS), which measure depression severity, were used to assess treatment response.	 	PMID:12082591
Clomipramine (Anafranil®)	ABCB1 (rs2032582) A>C (CC)	This patient's genetic profile suggests a reduced risk-involved response to clomipramine; blood plasma levels were used to assess treatment response.		PMID:21449676

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIDEPRESSANT

TCA

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Nortriptyline (Pamelor®)	SLC39A14 (rs17060812) C>T (CT)	This patient's genetic profile suggests a better response to nortriptyline; the Montgomery–Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:20970119
Nortriptyline (Pamelor®)	BDNF (rs962369) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation during nortriptyline treatment; the Hamilton Depression Rating Scale (HDRS-17), Beck Depression Inventory (BDI), and Montgomery-Åsberg Depression Rating Scale (MADRS), which measure depression severity and treatment response, were used to assess this risk.		PMID:19641488
Desipramine * (Norpramin®)	MC1R (rs2228478) A>G (AG)	This patient's genetic profile suggests a better response to desipramine; the measure of response was used to assess treatment efficacy.		PMID:21052032

STIMULANT

Amphetamine

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Dextroamphetamine (Dexedrine®)	DRD1 (rs4532) C>T (CT)	This patient's genetic profile suggests an increased risk of social withdrawal side effects when treated with dextroamphetamine; the Barkley side-effect scale, which measures the severity of side effects in children with ADHD, was used to assess treatment response.		PMID:23952191

CNS stimulant

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Methylphenidate (Ritalin®)	DRD1 (rs4532) C>T (CT)	This patient's genetic profile suggests an increased risk of social withdrawal side effects when treated with methylphenidate; the Barkley side-effect scale, which measures the severity of side effects in children with ADHD, was used to assess treatment response.		PMID:23952191

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

STIMULANT

CNS stimulant

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Methylphenidate (Ritalin®)	ABCB1 (rs2032582) A>C (CC)	This patient's genetic profile suggests a reduced risk of adverse events with methylphenidate; the Barkley Stimulant Side Effect Rating Scale (BSSERS), which measures stimulant side effects in children with ADHD, was used to assess potential side effects.		PMID:23771192
Methylphenidate (Ritalin®)	TH (rs2070762) A>G (AG)	This patient's genetic profile suggests a better response to methylphenidate; the Clinical Global Impression-Improvement scale, which measures treatment effectiveness, was used to assess treatment response.		PMID:26810137
Methylphenidate (Ritalin®)	CES1 (rs3815583) A>C (AC)	This patient's genetic profile suggests an increased risk of appetite suppression when treated with methylphenidate; the Barkley Stimulant Side Effect Rating Scale, which measures appetite reduction, was used to assess treatment response.		PMID:22688218
Methylphenidate (Ritalin®)	CYP2D6 (rs1135840) C>G (GG)	This patient's genetic profile suggests a reduced response to methylphenidate; the Conner's Parents Rating Scale-Revised (CPRS-R), which measures behavioral symptoms of ADHD, was used to assess treatment response.		PMID:36169235

ATYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Clozapine * (Clozaril®)	MTHFR (rs1801131) T>G (TG)	This patient's genetic profile suggests an reduced risk of weight gain when treated with clozapine; weight gain and metabolic parameters were monitored to assess treatment effects.		PMID:24725652
Olanzapine * (Zyprexa®)	MTHFR (rs1801131) T>G (TG)	This patient's genetic profile suggests an reduced risk of weight gain when treated with olanzapine; weight change and metabolic parameters were monitored to assess treatment effects.		PMID:24725652
Clozapine * (Clozaril®)	MTHFR (rs1801131) T>G (TG)	This patient's genetic profile suggests an reduced risk of weight gain when treated with clozapine; weight gain assessment was used to evaluate treatment response.		PMID:24725652
Olanzapine (Zyprexa®)	MTHFR (rs1801131) T>G (TG)	This patient's genetic profile suggests an reduced risk of weight gain when treated with olanzapine; weight change and metabolic parameters were monitored to assess treatment effects.		PMID:24725652

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ATYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Olanzapine (Zyprexa®)	SLC26A9 (rs11240594) G>A (AA)	This patient's genetic profile suggests a better response to olanzapine; a neurocognitive battery, which measures cognitive processing speed, was used to assess treatment response.		PMCID:PMC3055694
Clozapine * (Clozaril®)	GCG (rs13429709) T>C (TC)	This patient's genetic profile suggests an increased risk of weight gain while on clozapine; weight gain was monitored using percentage change from baseline weight.		PMID:24624910
Olanzapine * (Zyprexa®)	GCG (rs13429709) T>C (TC)	This patient's genetic profile suggests an increased risk of weight gain while on olanzapine; the percentage change from baseline weight was monitored to assess treatment effects.		PMID:24624910
Risperidone * (Risperdal®)	TMEFF2 (rs3738883) T>G (TG)	This patient's genetic profile suggests a better response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMCID:PMC3518380
Olanzapine (Zyprexa®)	UGT1A4 (rs2011425) T>G (TG) *1/*3A	This patient's genetic profile suggests a reduced risk of autonomic nervous system dysfunction when treated with olanzapine; power spectral analysis of heart rate variability, which measures the variation in time between heartbeats, was used to assess the patient's response.	 	PMCID:PMC7027321
Risperidone * (Risperdal®)	PPA2 (rs2636719) A>C (AC)	This patient's genetic profile suggests a reduced response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMCID:PMC3922978
Quetiapine * (Seroquel®)	PDE4D (rs2164660) G>A (GA)	This patient's genetic profile suggests a better response to quetiapine; the Patient Global Impression (PGI) and Clinical Global Impression-Severity (CGI-S), which measure treatment response and severity of illness, were used to assess treatment efficacy.		PMCID:PMC3922978
Risperidone (Risperdal®)	ADRB2 (rs1042713) G>A (AA)	This patient's genetic profile suggests an reduced risk of sexual adverse events when treated with risperidone; the UKU scale, which measures the severity of drug-related side effects, was used to assess treatment response.		PMCID:PMC3619141
Clozapine (Clozaril®)	DTNBP1 (rs742105) C>T (TT)	This patient's genetic profile suggests a better response to clozapine; the Positive and Negative Syndrome Scale score, which measures the severity of schizophrenia symptoms, was used to assess treatment response.		PMCID:PMC2857717

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ATYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Risperidone (Risperdal®)	CNR1 (rs1049353) C>T (CT)	This patient's genetic profile suggests an increased risk of weight gain with risperidone; weight gain and BMI change were monitored using standard clinical assessments.		PMCID:PMC3693401
Olanzapine (Zyprexa®)	PLAGL1 (rs3819811) A>G (GG)	This patient's genetic profile suggests a reduced response to olanzapine; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMCID:PMC3518380
Clozapine * (Clozaril®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to clozapine; the Positive and Negative Syndrome Scale (PANSS) negative subscale, which measures the severity of negative symptoms in schizophrenia, was used to assess treatment response.		PMID:27092952
Olanzapine * (Zyprexa®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to olanzapine; the Positive and Negative Syndrome Scale (PANSS) negative subscale, which measures the severity of negative symptoms in schizophrenia, was used to assess treatment response.		PMID:27092952
Quetiapine * (Seroquel®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to quetiapine; the Positive and Negative Syndrome Scale (PANSS) negative subscale, which measures the severity of negative symptoms in schizophrenia, was used to assess treatment response.		PMID:27092952
Risperidone * (Risperdal®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to risperidone; the Positive and Negative Syndrome Scale (PANSS) negative subscale, which measures the severity of negative symptoms in schizophrenia, was used to assess treatment response.		PMID:27092952
Risperidone (Risperdal®)	POR (rs1057868) C>T (CT) *1/*28	This patient's genetic profile suggests a moderate response to risperidone; blood plasma levels were monitored to assess treatment response.		PMID:26129906
Risperidone (Risperdal®)	POR (rs1057868) C>T (CT) *1/*28	This patient's genetic profile suggests a moderate response to risperidone; blood plasma levels and symptom scoring scales, which measure the severity of symptoms and treatment response, were used to assess the patient's condition.		PMID:26129906
Risperidone (Risperdal®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced risk of severe hyperprolactinemia when treated with risperidone; plasma levels for the parent compound risperidone, active metabolite, and prolactin were measured to assess treatment response.		PMID:29914302

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ATYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Aripiprazole * (Abilify®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a better response to aripiprazole; the measure of response was used to assess treatment efficacy.		PMID:36681895
Risperidone * (Risperdal®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced risk of weight gain during risperidone treatment; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMID:18718676
Risperidone (Risperdal®)	ABCB1 (rs1128503) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced response to risperidone; the Autism Treatment Evaluation Checklist (ATEC), which measures treatment effectiveness in autism, was used to assess treatment response.		PMID:19997080
Clozapine (Clozaril®)	SLC1A1 (rs2228622) G>A (GA)	This patient's genetic profile suggests an increased risk of developing obsessive-compulsive symptoms while receiving clozapine; allele, genotype, and haplotype frequencies were used to assess the potential for these symptoms.		PMID:19884611
Olanzapine (Zyprexa®)	SLC1A1 (rs2228622) G>A (GA)	This patient's genetic profile suggests an increased risk of obsessive-compulsive symptoms when treated with olanzapine; allele, genotype, and haplotype frequencies were used to assess the potential for adverse effects.		PMID:19884611
Risperidone (Risperdal®)	SLC1A1 (rs2228622) G>A (GA)	This patient's genetic profile suggests an increased risk of developing obsessive-compulsive symptoms when treated with risperidone; allele, genotype, and haplotype frequencies were used to assess the potential for adverse effects.		PMID:19884611
Olanzapine * (Zyprexa®)	CHAT (rs1880676) G>A (GA)	This patient's genetic profile suggests a better response to olanzapine; the Global Assessment of Function (GAF) scales and Positive and Negative Symptoms Scales (PANSS), which measure overall functioning and symptom severity in schizophrenia, were used to assess treatment response.		PMID:17503482
Clozapine (Clozaril®)	CYP2C18 (rs1126545) C>T (CT)	This patient's genetic profile suggests a modest increase in clozapine plasma levels; blood plasma levels were monitored to assess treatment response.		PMID:30922102
Risperidone * (Risperdal®)	BDNF (rs6265) C>T (CT)	This patient's genetic profile suggests a reduced risk of risperidone-induced weight gain; the Positive and Negative Syndrome Scale (PANSS), Extrapyramidal Symptom Rating Scale (ESRS), and UKU Side Effect Rating Scale, which measure psychiatric symptoms and side effects, were used to assess treatment response.		PMID:16633140

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ATYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Risperidone (Risperdal®)	BDNF (rs6265) C>T (CT)	This patient's genetic profile suggests an increased risk of risperidone-induced hyperprolactinemia; the Autism Treatment Evaluation Checklist (ATEC), prolactin levels, body mass index (BMI), and waist circumference were monitored to assess treatment response.		PMID:19997080
Iloperidone (Fanapt®)	CNTF (rs1800169) G>A (GA)	This patient's genetic profile suggests a reduced response to iloperidone; the Positive and Negative Syndrome Scale (PANSS), Brief Psychiatric Rating Scale (BPRS), and Clinical Global Impression (CGI), which measure psychiatric symptoms and overall clinical severity, were used to assess treatment response.		PMID:18303965
Iloperidone (Fanapt®)	ZFP91-CNTF (rs1800169) G>A (GA)	This patient's genetic profile suggests a reduced response to iloperidone; the Positive and Negative Syndrome Scale total score (PANSS-T), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMID:18303965
Risperidone (Risperdal®)	ANKK1 (rs1800497) G>A (AA)	This patient's genetic profile suggests a better response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to evaluate treatment response.		PMID:18855532
Aripiprazole (Abilify®)	ANKK1 (rs1800497) G>A (AA)	This patient's genetic profile suggests a better response to aripiprazole; the positive and negative syndrome scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMID:18926547
Clozapine * (Clozaril®)	HTR3A (rs1062613) T>C (CC)	This patient's genetic profile suggests a better response to clozapine; the Brief Psychiatric Rating Scale, which measures psychiatric symptoms severity, was used to assess treatment response.		PMID:20168265
Risperidone * (Risperdal®)	HTR2A (rs6313) G>A (AA)	This patient's genetic profile suggests a increased risk-involved response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMID:16633140
Risperidone (Risperdal®)	HTR2A (rs6311) C>T (TT)	This patient's genetic profile suggests a reduced risk-involved response to risperidone; the Autism Treatment Evaluation Checklist (ATEC), which measures the effectiveness of autism treatments, was used to assess treatment response.		PMID:19997080
Risperidone * (Risperdal®)	AKT1 (rs2494732) T>C (TC)	This patient's genetic profile suggests a reduced response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures the severity of symptoms in schizophrenia, was used to assess treatment response.		PMID:18855532

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ATYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Risperidone * (Risperdal®)	TJP1 (rs813676) T>C (TC)	This patient's genetic profile suggests a better response to risperidone; the Patient Global Impression (PGI) and Clinical Global Impression-Severity (CGI-S), which measure treatment response and severity of illness, were used to assess treatment efficacy.		PMCID:PMC3922978
Clozapine (Clozaril®)	SLC6A4 (rs25531) T>C (TC)	This patient's genetic profile suggests a better response to clozapine; the Brief Psychiatric Rating Scale (BPRS), which measures psychiatric symptoms severity, was used to assess treatment response.		PMID:20452607
Risperidone * (Risperdal®)	CCL2 (rs4586) T>C (TC)	This patient's genetic profile suggests a better response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures the severity of symptoms in schizophrenia, was used to assess treatment response.		PMID:24495780
Risperidone * (Risperdal®)	HRH4 (rs4483927) G>T (TT)	This patient's genetic profile suggests a reduced response to risperidone; the Positive and Negative Syndrome Scale (PANSS), which measures the severity of symptoms in schizophrenia, was used to assess treatment response.		PMID:23422377
Risperidone * (Risperdal®)	TNFRSF11A (rs2980976) G>A (GA)	This patient's genetic profile suggests a reduced response to risperidone; the Patient Global Impression (PGI) and Clinical Global Impression-Severity (CGI-S), which measure treatment response and severity of illness, were used to assess the effectiveness of the medication.		PMCID:PMC3922978
Olanzapine * (Zyprexa®)	GIPR (rs10423928) T>A (TA)	This patient's genetic profile suggests an increased risk of weight gain when treated with olanzapine; changes in BMI, which measure body weight relative to height, were monitored to assess treatment effects.		PMID:25321336
Risperidone * (Risperdal®)	COMT (rs9606186) C>G (CG)	This patient's genetic profile suggests a reduced response to risperidone; the Brief Psychiatric Rating Scale (BPRS), which measures psychiatric symptoms severity, was used to assess treatment response.		PMID:22935916
Quetiapine * (Seroquel®)	COMT (rs4818) C>G (CG)	This patient's genetic profile suggests a reduced response to quetiapine; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMID:26282453
Clozapine * (Clozaril®)	CYP3A5 *3/*3	This patient's genetic profile suggests a moderate response to clozapine; plasma concentrations of clozapine and N-desmethylclozapine were monitored to assess treatment levels.		PMID:27932669

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

NON-STIMULANT ADHD MEDICATION

NRI

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Atomoxetine (Strattera®)	HTR7 (rs7905446) T>G (GG)	This patient's genetic profile suggests a better response to atomoxetine; the 21-item Hamilton Depression Rating Scale (HDRS-21), which measures depression severity, was used to evaluate treatment response.		PMID:30874608

TYPICAL ANTIPSYCHOTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Chlorpromazine * (Thorazine®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to chlorpromazine; the Positive and Negative Syndrome Scale (PANSS) negative subscale, which measures the severity of negative symptoms in schizophrenia, was used to assess treatment response.		PMID:27092952
Trifluoperazine * (Stelazine®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to trifluoperazine; the Positive and Negative Syndrome Scale (PANSS), which measures symptom severity in schizophrenia, was used to assess treatment response.		PMID:27092952
Chlorpromazine * (Thorazine®)	CYP1A2 (rs762551) C>A (AA)	This patient's genetic profile suggests a reduced risk of acquiring a longer QT interval when treated with chlorpromazine; QT intervals measured by 12-lead electrocardiography (ECG) were monitored to assess cardiac function.		PMID:17611010

Butyrophenone

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Haloperidol * (Haldol®)	EPM2A (rs1415744) T>C (CC)	This patient's genetic profile suggests a better response to haloperidol; the Positive and Negative Syndrome Scale (PANSS) negative subscale, which measures the severity of negative symptoms in schizophrenia, was used to assess treatment response.		PMID:27092952

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

MOOD STABILIZER

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Lithium (<i>Lithobid</i> ®)	GSK3B (rs334558) A>G (AG)	This patient's genetic profile suggests a better response to lithium; treatment efficacy was evaluated for 83 patients.		PMID:23021822
Lithium (<i>Lithobid</i> ®)	FKBP5 (rs1360780) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation when treated with lithium; the Montgomery–Asberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676
Lithium (<i>Lithobid</i> ®)	HTR1B (rs130058) T>A (TA)	This patient's genetic profile suggests a better response to lithium; the Montgomery–Asberg Depression Rating Scale (MADRS), which measures depression severity, was used to assess treatment response.		PMID:21449676
Lithium (<i>Lithobid</i> ®)	ADCY1 (rs1521470) A>G (GG)	This patient's genetic profile suggests a better response to lithium; the measure of response was used to assess treatment efficacy.		PMCID:PMC5833535
Lithium (<i>Lithobid</i> ®)	ABCB1 (rs2032582) A>C (CC)	This patient's genetic profile suggests a reduced risk-involved response to lithium; the Montgomery–Asberg Depression Rating Scale (MADRS), which measures depression severity, was used to evaluate treatment response.		PMID:21449676
Lithium (<i>Lithobid</i> ®)	GRAMD1B (rs61123830) G>A (GA)	This patient's genetic profile suggests a reduced response to lithium; the measure of response was used to assess treatment efficacy.		PMCID:PMC5833535
Lithium (<i>Lithobid</i> ®)	CACNG2 (rs2284018) C>T (CT)	This patient's genetic profile suggests a better response to lithium; prospective data was used to assess treatment response.		PMID:18408563

ANTIEPILEPTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Ethosuximide * (<i>Zarontin</i> ®)	SCN1A (rs2298771) C>T (CT)	This patient's genetic profile suggests a better response to ethosuximide; seizure remission was used to assess treatment efficacy.		PMID:28753467
Levetiracetam * (<i>Keppra</i> ®)	SCN1A (rs2298771) C>T (CT)	This patient's genetic profile suggests a better response to levetiracetam; seizure remission was used to assess treatment response.		PMID:28753467

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIEPILEPTIC

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Oxcarbazepine * (Trileptal®)	SCN1A (rs2298771) C>T (CT)	This patient's genetic profile suggests a better response to oxcarbazepine; seizure remission was used to assess treatment response.		PMID:28753467
Cannabidiol (Epidiolex®)	ABCC5 (rs3749442) G>A (GA)	This patient's genetic profile suggests a reduced response to cannabidiol; the percent change in seizure frequency, which measures the effectiveness of treatment in reducing seizures, was used to assess treatment response.		PMID:34464454
Ethosuximide * (Zarontin®)	CACNA1H (rs61734410) C>T (CT)	This patient's genetic profile suggests a reduced response to ethosuximide; short-term seizure status was used to assess treatment effectiveness.		PMCID:PMC6171340
Ethosuximide * (Zarontin®)	CACNA1I (rs3747178) C>T (CT)	This patient's genetic profile suggests a reduced response to ethosuximide; short-term seizure outcomes were used to assess treatment effectiveness.		PMCID:PMC6171340

Mood stabilizer

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Valproic acid * (Depakote®)	LEPR (rs1137101) A>G (GG)	This patient's genetic profile suggests a reduced risk of weight gain when treated with valproic acid; body mass index (BMI) change was monitored to assess treatment effects.		PMCID:PMC4540110
Carbamazepine (Tegretol®)	EPHX1 (rs2234922) A>G (AG)	This patient's genetic profile suggests a moderate response to carbamazepine; the measure of response was used to assess the reduction in seizure frequency.		PMID:19620853
Lamotrigine * (Lamictal®)	SCN1A (rs2298771) C>T (CT)	This patient's genetic profile suggests a better response to lamotrigine; seizure remission was used to assess treatment response.		PMID:28753467
Carbamazepine * (Tegretol®)	GABRA1 (rs2279020) G>A (GA)	This patient's genetic profile suggests a reduced response to carbamazepine; blood plasma levels were monitored to assess treatment response.		PMCID:PMC3125053
Valproic acid * (Depakote®)	GABRA1 (rs2279020) G>A (GA)	This patient's genetic profile suggests a reduced response to valproic acid; blood plasma levels were monitored to assess treatment response.		PMCID:PMC3125053

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GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIEPILEPTIC

Mood stabilizer

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Carbamazepine * (Tegretol®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a better response to carbamazepine; plasma concentrations of carbamazepine and its major metabolites were monitored to assess treatment response.		PMCID:PMC4640545
Lamotrigine * (Lamictal®)	ABCB1 (rs1128503) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced response to lamotrigine; the LTG plasma concentration, which measures the level of the drug in the bloodstream, was used to assess treatment response.		PMID:32483200
Lamotrigine * (Lamictal®)	CYP3A4 (rs2740574) C>T (TT)	This patient's genetic profile suggests a reduced response to lamotrigine; drug concentrations in saliva and plasma were monitored to assess treatment response.		PMID:28343093
Valproic acid * (Depakote®)	CYP3A4 (rs2740574) C>T (TT)	This patient's genetic profile suggests a reduced response to valproic acid; drug concentrations in saliva and plasma were used to assess treatment response.		PMID:28343093
Valproic acid (Depakote®)	ANKK1 (rs1800497) G>A (AA)	This patient's genetic profile suggests a reduced risk of weight gain associated with valproic acid; the change in body mass index (BMI) was used to assess treatment response.		PMCID:PMC4540110
Valproic acid * (Depakote®)	GRIN2B (rs1019385) C>A (CA)	This patient's genetic profile suggests a moderate response to valproic acid; the maintenance dose and serum concentration were monitored to assess treatment effectiveness.		PMID:21806385
Carbamazepine * (Tegretol®)	CYP1A1 (rs2606345) C>A (CA)	This patient's genetic profile suggests an increased risk of seizures when treated with carbamazepine; seizure frequency, compliance to medications, adverse effects, and drug levels were monitored at 2, 4, 8, and 12 months.		PMID:21121773
Valproic acid * (Depakote®)	CYP1A1 (rs2606345) C>A (CA)	This patient's genetic profile suggests a increased risk-involved response to valproic acid; seizure frequency, compliance to medications, adverse effects, and drug levels were monitored using clinical assessments.		PMID:21121773
Carbamazepine * (Tegretol®)	CYP1A2 (rs762551) C>A (AA)	This patient's genetic profile suggests a better response to carbamazepine; blood plasma levels were monitored to assess treatment effectiveness.		PMID:26762380

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIEPILEPTIC

Mood stabilizer

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Lamotrigine * (Lamictal®)	CACNA1H (rs2753326) A>G (AG)	This patient's genetic profile suggests a better response to lamotrigine; EEG and clinical history were used to assess treatment effectiveness.		PMCID:PMC6171340
Valproic acid (Depakote®)	XBP1 (rs2269577) G>C (GC)	This patient's genetic profile suggests a better response to valproic acid; a scale described by Grof et al, which assesses prophylactic treatment response, was used to evaluate treatment response.		PMID:19564049
Valproic acid * (Depakote®)	CYP2C9 *1/*3	This patient's genetic profile suggests a reduced response to valproic acid; serum concentrations, dose, gender, age, and CYP2C9/2C19 genotypes were used to assess treatment response.		PMID:27353638

Hydantoin

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Phenytoin * (Dilantin®)	GABRA1 (rs2279020) G>A (GA)	This patient's genetic profile suggests a reduced response to phenytoin; blood plasma levels were monitored using standard therapeutic drug monitoring to assess treatment effectiveness.		PMCID:PMC3125053
Phenytoin * (Dilantin®)	CYP3A4 (rs2740574) C>T (TT)	This patient's genetic profile suggests a reduced response to phenytoin; drug concentrations in saliva and plasma were monitored to assess treatment response.		PMID:28343093
Phenytoin * (Dilantin®)	CYP1A1 (rs2606345) C>A (CA)	This patient's genetic profile suggests an increased risk of seizures when treated with phenytoin; seizure frequency, compliance to medications, adverse effects, and drug levels were monitored to assess treatment response.		PMID:21121773

Benzodiazepine

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Clobazam (Onfi®)	SCN1A (rs2298771) C>T (CT)	This patient's genetic profile suggests a better response to clobazam; seizure remission was used to assess treatment response.		PMID:28753467

Disclaimer: This report is intended to supplement a comprehensive patient assessment, including but not limited to diagnosis, clinical history, and evaluation of comorbid conditions. It should not be used as the sole basis for prescribing decisions. For the most current information on gene-drug associations, please refer to resources such as CPIC, FDA Table of Pharmacogenetic Associations. Please note that pharmacokinetics and pharmacodynamics may evolve over time, and this report may not reflect the latest findings. Prescribers should also consider approved drug labeling, clinical practice guidelines, and other relevant factors such as family history when making treatment decisions.

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIEPILEPTIC

Benzodiazepine

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Clobazam * (Onfi®)	POR (rs1057868) C>T (CT) *1/*28	This patient's genetic profile suggests a moderate response to clobazam; steady-state concentrations were monitored to assess the drug's effectiveness.		PMID:24345815

Barbiturate derivative

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Primidone * (Mysoline®)	CYP3A4 (rs2740574) C>T (TT)	This patient's genetic profile suggests a reduced response to primidone; drug concentrations in saliva and plasma were monitored to assess treatment response.		PMID:28343093

Barbiturate, depressant

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Phenobarbital (Luminal®)	ABCB1 (rs1045642) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced response to phenobarbital; blood plasma levels were monitored to assess treatment effectiveness.		PMID:25846690
Phenobarbital (Luminal®)	CYP1A1 (rs2606345) C>A (CA)	This patient's genetic profile suggests an increased risk of seizures when treated with phenobarbital; seizure frequency, compliance to medications, adverse effects, and drug levels were monitored to assess treatment response.		PMID:21121773

ANTICONVULSANT

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Perampanel * (Fycompa®)	CYP3A5 *3/*3	This patient's genetic profile suggests a reduced response to perampanel; plasma concentrations were measured by high-performance liquid chromatography with ultraviolet detector.		PMID:38381330

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

OPIOID ANALGESIC

Depressant

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Morphine (MS Contin®)	BDNF (rs6265) C>T (CT)	This patient's genetic profile suggests a increased risk-involved response to morphine; blood plasma levels and symptom scoring scales, which measure the severity of symptoms, were used to assess treatment response.		PMCID:PMC11144121

STATIN

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Atorvastatin (Lipitor®)	ABCB1 (rs1128503) A>G (GG) *1/*1	This patient's genetic profile suggests a reduced risk of atorvastatin toxicity; CK activity measurement and genotyping were used to assess the risk of statin-related myopathy.	 	PMCID:PMC10587173

THYROID HORMONE REPLACEMENT

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Liothyronine (Cytomel®)	FKBP5 (rs1360780) T>C (TC)	This patient's genetic profile suggests an increased risk of suicidal ideation when treated with liothyronine; the MADRS suicidal item scale, which measures suicidal thoughts and behaviors, was used to assess this risk.		PMID:21449676
Liothyronine (Cytomel®)	HTR1B (rs130058) T>A (TA)	This patient's genetic profile suggests a better response to liothyronine; the 10th item on the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures suicidal ideation, was used to assess treatment response.		PMID:21449676
Liothyronine (Cytomel®)	ABCB1 (rs2032582) A>C (CC)	This patient's genetic profile suggests a reduced risk-involved response to liothyronine; the Montgomery-Åsberg Depression Rating Scale (MADRS), which measures depression severity, was used to evaluate treatment response.		PMID:21449676

GENE DRUG INTERACTION

Drug Synopsis with risk assessment

ANTIVIRAL

Nucleoside analogue

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Ribavirin (Copegus®)	IL6 (rs1800795) C>G (GG)	This patient's genetic profile suggests a increased risk-involved response to ribavirin; the Beck Depression Inventory and the Zung Self-Rating Depression Scale, which measure symptoms of depression, were used to assess treatment response.		PMCID: PMC3513412

IMMUNOMODULATOR

Interferon

Drug	Gene / Genotype	Clinical Impact	PGx Response	Citations
Peginterferon alfa-2b (PegIntron®)	IL6 (rs1800795) C>G (GG)	This patient's genetic profile suggests an increased risk-involved response to peginterferon alfa-2b; the Beck Depression Inventory and the Zung Self-Rating Depression Scale, which measure symptoms of depression, were used to assess treatment response.		PMCID: PMC3513412

ADDITIONAL INFORMATION

TEST METHODOLOGY

1. Genes Analysed: Sequencing of the coding regions (CDS) of 19,433 genes was carried out. Analysis of genes relevant to the condition is performed to identify potential genetic alterations.
2. Quality Metrics - Total Reads: 81,295,668, Total Mapped Reads: 81,003,478, Percent on Target Reads: 99.64%, Average Unique Coverage (Count): 255.47x. It varies by patient to patient.
3. Sample Preparation: DNA was extracted using commercial buccal swab collection kits from samples obtained via non-invasive buccal swabbing. The extracted DNA was subjected to multiple quality assessment steps to ensure its suitability for sequencing. Internal validation confirmed that all samples met established laboratory quality control (QC) standards, including both sample QC and library QC. Only those samples that passed the QC criteria proceeded to the library preparation stage.
4. Library Preparation and Assay Sequencing: Libraries were prepared using enzymatic shearing and ligation-based library preparation, following the library preparation protocols provided by Twist Biosciences (Library Preparation EF 2.0 with Enzymatic Fragmentation and Twist Universal Adapter System). Subsequently, the libraries were subjected to capture and multiplexing processes in accordance with Twist capture protocol using Twist Exome 2.0 + Comp. Exome Spike-In NGS Panel. After the quality check, the enriched libraries were then subjected to sequencing on an Illumina NovaSeq instrument, using 2 x 150 base paired end reads for analysis.
5. Quality Assessment: Nucleic acid quality was checked using Qubit fluorometry and Agilent TapeStation. QC was performed after extraction, fragmentation, indexing, and library amplification. Only samples meeting QC standards proceeded further.
6. Variant Calling: Sequencing data underwent alignment, post-processing, and quality filtering using the Illumina DRAGEN Bio-IT platform. The DRAGEN v4.2.4 pipeline was used to align reads against the GRCh38 (hg38) human reference genome. This process included steps such as read mapping, duplicate marking, base quality recalibration, and variant calling. The resulting output contained high-confidence genetic variants derived from the whole exome data.
7. Data Mapping and Reporting: The resulting VCF files containing variant information were then mapped against the POD database using key fields. The mapped variants were subsequently reviewed by a pathologist, and a final clinical report was generated based on the validated findings.

DISCLAIMER

The content provided as a professional service by POD Health has not been reviewed or approved by the U.S. Food and Drug Administration (FDA). This diagnostic report, including all findings and interpretations, is intended solely for use by qualified medical professionals. It is based on the analysis of a buccal swab sample obtained from the patient and processed through Whole Exome Sequencing (WES), which was conducted at **Karkinos Healthcare** certified genomics facility using standardized protocols to ensure high-fidelity variant detection. The resulting sequencing data was subsequently analyzed using curated pharmacogenomic databases to identify clinically relevant gene drug interactions tailored to the patient's genetic profile. The individual from whom the sample was collected has been clinically diagnosed with a neurological, neurodegenerative, or neurodevelopmental disorder. This report is intended to function as a companion diagnostic tool to assist in personalized treatment planning and clinical decision-making.

This report is intended for medical and scientific evaluation purposes only. ALL methodologies, including the WES technology and pharmacogenomic interaction analysis utilized in this report, are currently being validated as part of a clinical study and have not been certified by the National Accreditation Board for Testing and Calibration Laboratories (NABL). While these methodologies are considered highly reliable, no diagnostic procedure is 100% accurate. Variations in test results may occur due to factors such as sample quality, genetic variability, or other clinical considerations.

The information contained in this report is intended to support, not replace, the professional judgment of the treating neurologist or medical professional. The presence or absence of ANY specific variant or pharmacogenomic interaction should be considered alongside other clinical and diagnostic findings when formulating the appropriate treatment plan for the patient. The ordering medical professional is responsible for interpreting these findings within the context of the patient's overall medical condition.

By using the information provided in this report, the ordering medical professional acknowledges that any decisions made based on the diagnostic findings are at their sole discretion. POD Health, Inc., along with its associated laboratories and affiliates, shall not be liable for any interpretations or treatment decisions made based on this report.

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